



Q1. What is the difference between nominal voltage, Voc, Vmp, Isc and Imp in case of a solar panel? Which parameters are important to check before installation of solar panels?

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A1. Different solar panels or photovoltaic (PV) modules have different specifications. There are several terms associated with solar panels and their ratings such as nominal voltage, Voc, Vmp, Isc and Imp. All these parameters are checked before installation. Brief information about each is given below.

Nominal voltage. Used just for classification, it is not a real voltage you are going to measure. It is not a fixed voltage either and, normally, it is not mentioned in the specification sheet of a PV module. Some of the common parameters mentioned in the specification sheet are listed in the table.

Voltage at open circuit (Voc). This voltage is checked with a voltmeter across the output terminals of the solar panel module, without connecting any load. This parameter is used to check/test the module during installation and later for system design. It is an important parameter under standard test conditions. Voc is used while determining the number of solar panels required for a particular load.

Voltage at maximum power (Vmp). This is the voltage available when the panel is connected to a load and is operating at its maximum capacity under standard test conditions. Most solar panel manufacturers specify Vmp

Typical Specification of a 250W Solar Panel

Electrical parameters	Typical value
Power Max (Pm)	250
Short Circuit Current (Isc)	8.95A
Max Power Current (Imp)	8.35A
Maximum Voltage (Vmp)	29.95
Open Circuit Voltage (Voc)	37.25
System Voltage	1000V DC

to be around 70 to 80 per cent of the Voc.

Short circuit current (Isc). This is the value of current obtained when the positive and negative terminals of the panel are connected to each other through an ammeter in series. This is the highest current the solar panel will produce under standard test conditions. Isc is used to determine how many amps a panel can handle when connected to a device like a solar charge controller or an inverter circuit.

Current at maximum power (Imp). This current is obtained when solar panel is producing its maximum power. It is the amperage you would want to see when connected to a solar equipment.

Q2. How does the prepaid smart energy meter work? What are some of its advantages?

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A2. A smart energy meter is used to measure electrical current flow and voltage at regular intervals. It calculates the energy consumption and shows it on the display of the meter installed at your home. It sends this information to your power/energy supplier also using a communication link like wide area network (WAN).

In a prepaid smart energy meter, the consumer buys a recharge card and pays for the energy units required by him in advance, like the prepaid connection for a mobile phone. The balance amount keeps reducing with each unit of energy consumed. If the balance amount becomes zero, the power supply automatically gets cut off. When you top-up your prepaid card, the balance increases accordingly and the relay (an automatic switch) allows power to start

flowing once again.

There are many advantages of a prepaid smart energy meter. It allows you to calculate your daily usage and manage your account via a smartphone app or web portal. As consumer, you can view your current balance and estimated usage for the days ahead.

You will be notified via text message or email about daily energy consumption and your existing balance. This will help you to know when to recharge the prepaid card before the balance becomes zero. The display of the prepaid energy meter also shows your energy consumption and the balance amount.

The meter shows energy consumption in kilo-watt-hours (or kWh), where one kWh refers to the unit of energy required to provide 1000 watts of power for one hour.